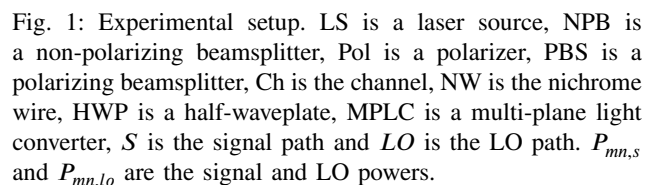


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I. INTRODUCTION

II. EXPERIMENTAL RESULTS

To quantify relative WFEs, we take the difference ΔP_{mn} between the normalized signal power $P_{mn,s}/P_s$ and the LO power $P_{mn,lo}/P_{lo}$ in each mode, $\Delta P_{mn} = (P_{mn,s}/P_s) - (P_{mn,lo}/P_{lo})$. If the wavefronts of the signal and the LO are the same, then $P_{mn,s}/P_s = P_{mn,lo}/P_{lo}$, so $\Delta P_{mn} = 0$, as is commonly assumed.



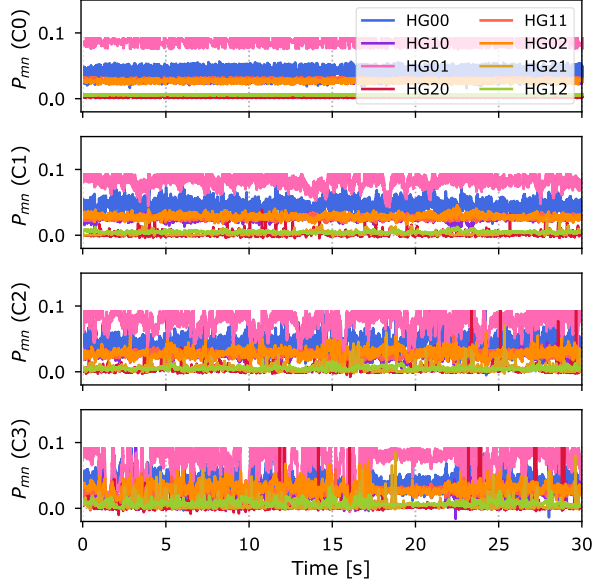


Fig. 2: Time versus ΔP_{mn} for turbulence Cases 0-3 (where C represents Case).

Otherwise, there are relative WFEs.

Time is plotted against ΔP_{mn} for each of the turbulence cases, over a period of 30 s, in Fig. 2. It can be immediately seen that *relative WFEs are present* between the signal and the LO ($\Delta P_{mn} > 0$) for all cases. Given that relative WFEs are present in Case 0, they are likely caused by imperfections or inaccurate calibration of optical hardware [2]. It can be also seen in Fig. 2 that ΔP_{mn} is highest for the HG₀₁ mode, followed by the HG₀₀ mode, then the HG₁₀, HG₁₁, and HG₀₂ modes (layered on top of each other). We also see that the HG₂₀, HG₂₁, and HG₁₂ modes approach zero (layered on top of each other). In general, the fluctuations in ΔP_{mn} for each mode remain relatively constant for Case 0, indicating that the WFEs are relatively constant.

From Fig. 2, the same ordering of the modes, from the lowest to the highest ΔP_{mn} values, is found for the turbulent Cases 1-3. However, it can be seen that the ΔP_{mn} fluctuations become noisier as the turbulence strength increases, which we quantify in Fig. 3 using the variance of ΔP_{mn} . Although the initial WFEs are likely caused by imperfections or imprecise calibration of the optical hardware, the increase in variances of ΔP_{mn} across all modes, as turbulence increases, indicates that the WFEs are affected by the turbulence itself. In addition, we find that the WFEs in each mode are statistically significant by calculating a two-sample Kolmogorov–Smirnov test [5] using the distributions $P_{mn,s}/P_s$ and $P_{mn,lo}/P_{lo}$. Our results show that the differences in all distributions are statistically significant, with a 99.99% confidence, which supports the existence of relative WFEs.

We note that our experimental CV-QKD setup may not

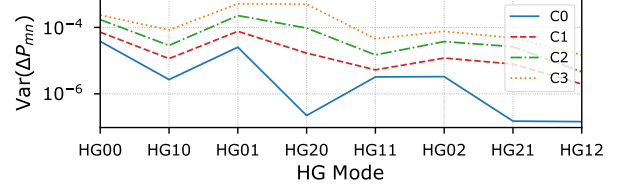


Fig. 3: HG mode versus ΔP_{mn} variance for turbulence Cases 0-3 (where C represents Case).

represent all CV-QKD setups, as we adopt path differences for the LO and signal, and apply a small wavelength perturbation for co-measurement of the signal and LO using the MPLC. However, our experiments highlight how relative WFEs may occur in real CV-QKD setups. This result is fundamental to CV-QKD across atmospheric channels; if imperfections in optical hardware and turbulence cause fluctuating relative WFEs, then correcting for them could prove vital in attaining positive key rates across free-space channels. The deleterious effects of relative WFEs on free-space CV-QKD would be particularly pronounced in the future satellite-Earth links required for a global Quantum Internet. To mitigate the effect of WFEs in CV-QKD, our ongoing work proposes a machine learning-based solution to correcting WFEs [2], which can lead to lower excess noise and higher key rates across atmospheric channels.

III. CONCLUSION

We have experimentally demonstrated the presence of relative wavefront errors between a polarization-multiplexed strong classical local oscillator and a weak “quantum” signal after passing both through turbulent channels. In addition, we demonstrated how the turbulence strength affects these relative wavefront errors. The presence of relative wavefront errors in CV-QKD setups has the potential to negatively impact QKD secure key rates, necessitating the development of wavefront correction methods to establish a global CV-QKD network.

REFERENCES

- [1] E. Villaseñor *et al.*, “Atmospheric effects on satellite-to-ground quantum key distribution using coherent states,” in *Global Communications Conference*, IEEE, 2020.
- [2] N. K. Long *et al.*, “Quantum wavefront correction via machine learning for satellite-to-Earth CV-QKD,” *in prep.*, 2025.
- [3] V. Billault *et al.*, “Free space optical communication receiver based on a spatial demultiplexer and a photonic integrated coherent combining circuit,” *Optics Express*, vol. 29, no. 21, pp. 33 134–33 143, 2021.
- [4] DeLange, O.E., “Optical heterodyne detection,” *IEEE Spectrum*, vol. 5, no. 10, pp. 77–85, 1968.
- [5] J. W. Pratt *et al.*, “Kolmogorov-Smirnov two-sample tests,” *Concepts of Nonpar. Theory*, pp. 318–344, 1981.